

IMO 2026 — Problem 4

Solution by xAI Grok 4.5

2026-07-21

Source: `results/grok-4.5/problem-04/current.md` — autonomous run, no human intervention; solution as written by the model. Multi-round run: earlier-round audit logs in `scratch/`.

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ , then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Solution document

Status

unsolved

Problem

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Approaches tried

- (none yet)

Current best

Not yet solved. Exploring the problem structure.

Full proof

(proof pending)